



```

super@ubuntu:~$ gcc sctpserver.c -lsctp -o server
super@ubuntu:~$ ./server
Server Running
Connected to 127.0.0.1
Sent: Message #1

Sent: Message #2

Sent: Message #3

Server Running

```

Figure 1. Server output

```

char buffer[MAX_BUFFER+1] = "Message ##\n";
sfd = socket( AF_INET, SOCK_STREAM, IPPROTO_SCTP );
bzero( (void *)&saddr, sizeof(saddr) );
saddr.sin_family = AF_INET;
saddr.sin_addr.s_addr = htonl( INADDR_ANY );
saddr.sin_port = htons(29008);
bind( sfd, (struct sockaddr *)&saddr, sizeof(saddr) );
/* Maximum of 3 streams will be available per socket */
memset( &initmsg, 0, sizeof(initmsg) );
initmsg.sinit_num_ostreams = 3;
initmsg.sinit_max_instreams = 3;
initmsg.sinit_max_attempts = 2;
setsockopt( sfd, IPPROTO_SCTP, SCTP_INITMSG,
            &initmsg, sizeof(initmsg) );
listen( sfd, 5 );

for(;;) {
    printf("Server Running\n");
    len=sizeof(caddr);
    cfd=accept(sfd, (struct sockaddr *)&caddr, &len);
    printf("Connected to %s\n",
           inet_ntop(AF_INET, &caddr.sin_addr, buff,
                     sizeof(buff)));
    for(i=0; i< 3; i++) {
        /* Changing 9th character the character after # in the
        message buffer */
        buffer[9] = '1'+i;
        sctp_sendmsg( cfd, (void *)buffer, (size_t)
        strlen(buffer),
                     NULL, 0, 0, 0, 1 /* stream */, 0, 0 );
        printf("Sent: %s\n", buffer);
    }
    close( cfd );
}
return 0;
}

```

And here's the code of the client, *sctpclient.c*:

```

#include <stdio.h>
#include <string.h>

```



```

super@ubuntu:~$ ./client 127.0.0.1
Received following data on stream 0
Message #1

Received following data on stream 1
Message #2

Received following data on stream 2
Message #3

$

```

Figure 2. Client output

```

#include <sys/socket.h>
#include <sys/types.h>
#include <netinet/in.h>
#include <netinet/sctp.h>

#define MAX_BUFFER 1024

int main(int argc, char **argv) {
    int cfd, i, flags;
    struct sockaddr_in saddr;
    struct sctp_sndrcvinfo sndrcvinfo;
    struct sctp_event_subscribe events;
    struct sctp_initmsg initmsg;
    char buffer[MAX_BUFFER+1];

    if(argc!=2) {
        printf("Usage: %s ipaddress\n", argv[0]);
        return -1;
    }

    cfd = socket( AF_INET, SOCK_STREAM, IPPROTO_SCTP );
    /* Maximum of 3 streams will be available per socket */
    memset( &initmsg, 0, sizeof(initmsg) );
    initmsg.sinit_num_ostreams = 3;
    initmsg.sinit_max_instreams = 3;
    initmsg.sinit_max_attempts = 2;
    setsockopt( cfd, IPPROTO_SCTP, SCTP_INITMSG,
                &initmsg, sizeof(initmsg) );
    bzero( (void *)&saddr, sizeof(saddr) );
    saddr.sin_family = AF_INET;
    inet_pton(AF_INET, argv[1], &saddr.sin_addr);
    saddr.sin_port = htons(29008);

    connect( cfd, (struct sockaddr *)&saddr, sizeof(saddr) );
    memset( (void *)&events, 0, sizeof(events) );
    events.sctp_data_io_event = 1;
    setsockopt( cfd, SOL_SCTP, SCTP_EVENTS,
                (const void *)&events, sizeof(events) );
    /* Sending three messages on different streams */
    for (i=0; i<3; i++) {
        bzero( (void *)&buffer, sizeof(buffer) );
        sctp_rcvmsg( cfd, (void *)buffer, sizeof(buffer),
                    (struct sockaddr *)NULL, 0, &sndrcvinfo,

```